



CLOUDPANDITH ONLINE TRAININGS



Master Data Engineering with Live Training
Azure Data Factory | Databricks | Microsoft Fabric

- ◆ Live Classes
- ◆ Real-Time Projects
- ◆ Interview Preparation
- ◆ Lifetime Access

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1. Delta Lake Architecture

SQL DW → Structured Data, ACID, CRUD

ACID Properties (Simple + Real-Time Understanding)

● 1. Atomicity (All or Nothing)

👉 Meaning: A transaction must complete fully or not at all.

Example:

A has ₹50K

B has ₹20K

Transfer ₹10K from A → B

Steps:

Deduct from A → ₹40K ✓

Add to B → ₹30K ✓

✓ Final state is correct

✗ But imagine failure:

A deducted → ₹40K

B NOT credited ✗

👉 Money is lost → system failure

✓ So DB does this:

If ANY step fails → rollback everything

A goes back to ₹50K

👉 That's Atomicity

● 2. Consistency (Data Should Always Be Correct)

👉 Meaning: Data must follow rules before and after transaction.

Rule here: Total money should remain same

Before transaction:

A = ₹50K

B = ₹20K

👉 Total = ₹70K

Correct After transaction:

A = ₹40K

B = ₹30K

👉 Total = ₹70K ✓

Wrong scenario:

A = ₹50K

B = ₹30K

👉 Total = ₹80K ❌

👉 System created extra money → inconsistent

✓ DB ensures:

No rule violations

No duplicate / missing data

👉 That's Consistency

● 3. Isolation (No Interference Between Transactions)

👉 Meaning: Multiple transactions should not affect each other.

Example:

Transaction 1:

A → B (₹10K)

Transaction 2:

C → D (₹10K)

👉 Both run at same time

❌ Problem if no isolation:

T1 reads half-updated data from T2

Dirty / incorrect values

✓ With isolation:

Each transaction behaves like it's running alone

Only sees committed data

👉 No confusion, no overlap

👉 That's Isolation

● 4. Durability (Data is Permanent)

👉 Meaning: Once transaction is committed → it is saved permanently.

Example:

Transfer completed

A = ₹40K, B = ₹30K

⚡ Suddenly:

Power failure / system crash

✗ Without durability:

Data lost

✓ With durability:

Data stored in logs + disk

System restarts → restores committed data

👉 Because:

Before confirming transaction → DB writes to transaction logs

👉 That's Durability

Problems with SQL DW

1. Only structured data
2. Costly
3. Fixed schema

2. Data Lake

Supports structured, semi-structured, unstructured data

Cheap storage

No ACID → Data corruption risk

3. Delta Lake

Supports all data types

Cheap

ACID + CRUD supported

Schema evolution supported

Storage formats:

CSV

Parquet

Delta (with transaction logs)

4. SQL DW vs Data Lake vs Delta Lake

Feature	SQL DW	Data Lake	Delta Lake
Data Type	Structured	All types	All types
Cost	High	Low	Low
ACID	Yes	No	Yes
Schema	Fixed	Flexible (unsafe)	Flexible (safe)
CRUD	Yes	Limited	Yes
Use Case	Reporting	Storage	Analytics + ETL

5. Delta Lake Internals

Delta uses transaction logs (`_delta_log`)

Each operation creates new version logs (`0.json, 1.json...`)

Example Flow:

Step1: Table created → log 0

Step2: Schema change → log 1

Step3: Insert → log 3

Files stored in parquet + logs

Old files are marked as removed but not deleted immediately.

VACUUM cleans files after 7 days.